Theorems:

Week 1: **Theorem 1.1** Let A and B be statements. Then A $\Rightarrow$ B is equivalent to $\overline{\text{A}}\vee$B. That is, (A$\Rightarrow$B) $\iff$ ($\overline{\text{A}}$ $\vee$B).

Theorem 1.2 Let A and B be statements. Then A $\Rightarrow$ B is equivalent to $\overline{\text{B}} \Rightarrow \overline{\text{A}}$. That is, (A $\Rightarrow$ B) $\iff$ ($\overline{\text{A}}\vee$B).

Theorem 1.3 Let A and B be statements. Then $\overline{\text{A} \vee \text{B}}$ is equivalent to $\overline{\text{A}} \wedge \overline{\text{B}}$.

Theorem 2.4 Given any two real numbers a $<$ b, there is some r $\in$Q such that a $<$r $<$ b.

Theorem 2.5 The number $\sqrt{2}$ is irrational.

Week 2:

Theorem 3.7 -AM-GM Inequality Let n$\in$N and $x_1, \ldots, x_n \geq 0$. Then $(x_1 \cdot x_2 \ldots x_n)^{1/n} \leq \frac{(x_1+\ldots+x_n)}{n}$

Theorem 3.8 -Cauchy-Schwarz Inequality Let n $\in$ N, $x_1,\ldots,x_n$, y_1, ..., y_n $\in$ R. Then $x_1y_1 + \cdots + x_ny_n \leq$ $\sqrt{x_1^2+\ldots+x_n^2}\,\sqrt{y_1^2+\ldots+y_n^2}$

Theorem 3.8 Triangle Inequality For x,y $\in$ R, $|x + y| \leq |x| + |y|$.

Corollary 3.12 Reverse Triangle Inequality $||x|-|y|| \leq |x+y|$. in particular: $|x|-|y| \leq |x+y|$

Theorem 4.1 -Geometric Series Let x $\neq$ 1 be a real number. Then for N $\in$ N, $\sum_{n=1}^{N} x^n = \frac{x-x^{N+1}}{1-x}$ and $\sum_{n=0}^{N} x^n = \frac{1-x^{N+1}}{1-x}$

Theorem 4.2 Every Integer n $\geq$ 2 can be written as a product of primes n = $p_1 \ldots p_k$

Week 3: **Completeness Axiom for the real numbers:** Every nonempty subset of R that is bounded above has a least upper bound.

Theorem 5.1 he Archimedean property holds. That is, for any x, y $\in$ R with x, y $>$ 0, there is some n $\in$ N such that ny $>$ x.

Theorem 6.6 Let x $>$ 0 and k $\in$ N. Then there is a unique y $>$ 0 such that y^k = x.

Week 4: **Theorem 7.1 Monotone Convergence Theorem** Let (a_n) be an increasing sequence of real numbers that is bounded above (i.e. there is M so that $a_n \leq$ M for all n). Then (a_n) converges to some limit. If (a_n) is a decreasing sequence and is bounded below, then (a_n) converges to some limit.

Theorem 8.1 Given a sequence (a_n) with $a_n \in$ 0, 1, . . . , 9 for all n, the decimal expansion x = $0.a_1a_2 \ldots a_n \ldots$ given by the limit of the sequence $x_n = \frac{a_1}{10} + \frac{a_2}{100} + \ldots + \frac{a_n}{10^n}$ defines a real number x satisfying $0 \leq x \leq 1$

Theorem 8.2 Let x $\in$ R satisfy 0 $\leq$ x $<$ 1. Then there is a sequence of integers (a_n), with $a_n \in$ 0, 1, . . . , 9 for all n, so that if we let $x_n = \frac{a_1}{10} + \frac{a_2}{100} + \ldots + \frac{a_n}{10^n}$ then $x_n \to$ x as n $\to \infty$

Theorem 8.3 Let 0 $\leq$ x $<$ 1 and suppose that x = $0.a_1a_2\ldots = 0.b_1b_2\ldots$. Let ℓ be the smallest integer for which $a_\ell \neq b_\ell$ and suppose that $a_\ell < b_\ell$. Then $b_\ell = a_\ell + 1$, and for all k $>$ ℓ we have b_k = 0 and a_k = 9

Lemma 8.4 if x $\geq$ 0 is rational, then it has a periodic decimal expansion, that is, x = $a_0.a_1.a_2 \ldots a_k\overline{b_1b_2\ldots b_\ell}$

Corollary 8.5 If p and q are positive integers, the rational number p/q has a decimal expansion with period at most q.

Lemma 8.6 If x $\geq$ 0 has a periodic decimal expansion, then x is rational.

Theorem 8.7 A number x $\geq$ 0 is rational $\iff$ it has a periodic decimal expansion.

Theorem 8.11 -Comparison Test If $b_j \geq 0$ for all j and $\sum_{j=1}^{\infty} b_j$ is convergent, and if $|a_j| \leq b_j$ for all j, then $\sum_{j=1}^{\infty} a_j$ is also convergent, and moreover $|\sum_{j=1}^{\infty} a_j| \leq \sum_{j=1}^{\infty} b_j$